

TECHNICAL DOCUMENTATION: ALGORITHMIC FAIRNESS, CLASSIFICATION, AND CLUSTERING

Lead Analyst & Author: Mohammed Faham

*Dataset Domain: Sociological Recidivism Risk (COMPAS -
propublica_data_for_fairml.csv)*

*Target Variable: Two_yr_Recidivism (Binary: 0 = No, 1 =
Yes)*

1. Data Preprocessing & Mathematical Scaling

The dataset comprises 6,172 observations across 11 socio-economic and demographic features with zero missing values. Categorical features (e.g., gender, race, misdemeanor charges) were natively one-hot encoded into binary integers (0 or 1).

Feature Standardization

Distance-based algorithms (K-Means, KNN, SVM) calculate geometric proximity between data points. Features with larger numerical ranges, such as Number_of_Priors (ranging from 0 to 38), inherently dominate the distance calculations over binary demographic features (ranging from 0 to 1). To neutralize this magnitude bias, StandardScaler was applied to center the feature columns around a mean of zero with a standard deviation of one.

The transformation applied to every scalar value x in a feature column is:

$$z = \frac{x - \mu}{\sigma}$$

Where μ is the mean of the training samples and σ is the standard deviation. To prevent data leakage during supervised learning, the scaler was strictly fitted to the training split (80%) and subsequently applied to transform both the training and testing sets.

2. Unsupervised Learning: K-Means Clustering & Dimensionality Reduction

The primary objective of the unsupervised phase was to determine if the continuous and binary features naturally segregate individuals into distinct risk profiles without exposing the algorithm to the target label.

2.1 Optimization of k (Cluster Count)

The K-Means algorithm partitions the n observations into k sets by minimizing the Within-Cluster Sum of Squares (WCSS), also known as Inertia:

$$WCSS = \sum_{j=1}^k \sum_{x_i \in C_j} ||x_i - \mu_j||^2$$

To mathematically justify the optimal number of clusters, both the Elbow Method (tracking WCSS) and the Silhouette Score were calculated for $k \in [2,10]$. The Silhouette Score measures how similar an object is to its own cluster compared to other clusters:

$$S(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

Where $a(i)$ is the mean intra-cluster distance and $b(i)$ is the mean nearest-cluster distance.

- **Findings:** The algorithm identified $k = 2$ as the most stable macro-level grouping (Inertia = 59,372; Silhouette = 0.163). While higher values of k artificially inflated the Silhouette Score, $k = 2$ provided the most analytically relevant partition for the binary nature of recidivism risk.

2.2 Unsupervised Demographic Profiling

Applying $k = 2$ resulted in two mathematical clusters that closely mirrored the actual recidivism risk:

- **Cluster 0 (High-Risk Geometric Group):** Displayed a true recidivism rate of 62.9% and an average of 5.2 prior offenses. Crucially, this cluster was 72.7% African American.
- **Cluster 1 (Low-Risk Geometric Group):** Displayed a true recidivism rate of 31.0% and an average of 1.5 prior offenses. This cluster was 33.6% African American.
- **Conclusion:** The unsupervised distance calculations isolated a severe structural bias. The algorithm intrinsically clustered data along racial lines because proxy variables (like criminal history) possess high collinearity with demographic indicators in the underlying data.

2.3 Dimensionality Reduction (PCA)

To address the low Silhouette Score (0.163) caused by the sparse 11-dimensional space, Principal Component Analysis (PCA) was implemented. Instead of hardcoding dimensions, the algorithm was mathematically constrained to retain 95% of the dataset's variance.

- **Results:** PCA retained 10 of the 11 original features to meet the 95% threshold. This confirmed incredibly low multicollinearity across the dataset—nearly every socio-economic variable provided unique variance. Consequently, the dimensionally reduced Silhouette Score only improved marginally to 0.1663, validating that compressing this specific dataset into arbitrarily fewer dimensions (e.g., 3 dimensions for visualization) inherently destroys critical predictive data.

3. Supervised Learning: Classification Modeling

To predict the `Two_yr_Recidivism` target, two mathematically distinct baseline algorithms were evaluated and subsequently optimized using 5-fold cross-validated grid search (`GridSearchCV`).

3.1 K-Nearest Neighbors (KNN) Optimization

The KNN algorithm classifies an observation by a majority vote of its neighbors. The grid search evaluated multiple values for k alongside Euclidean and Manhattan distance metrics.

- **Manhattan Distance:**

$$d(x, y) = \sum_{i=1}^n |x_i - y_i|$$

- **Grid Search Results:** The optimal parameters rejected the default $k = 5$ and Euclidean distance, selecting $n_neighbors = 25$, $weights = 'uniform'$, and $metric = 'manhattan'$. The Manhattan grid-based metric proved superior in navigating the orthogonal binary features (like demographic flags) alongside the continuous variables.

3.2 Support Vector Machine (SVM) Optimization

The SVM algorithm constructs hyperplanes in a multidimensional space to separate classes. Because sociological data is rarely linearly separable, the Radial Basis Function (RBF) kernel was applied to project the data into an infinite-dimensional space.

- **RBF Kernel:**

$$K(x, x') = \exp(-\gamma ||x - x'||^2)$$

- **Grid Search Parameters:** Tested the regularization parameter C (penalty for misclassification) and γ (radius of influence for support vectors).
- **Grid Search Results:** The optimal parameters selected were $C = 50$ and $\gamma = 'scale'$. The high C value forced a strict penalty on margin violations, creating a highly complex and tightly fitted decision boundary.

4. Final Evaluation & Algorithmic Fairness Audit

Both optimized models reached a mathematically identical performance ceiling, confirming that maximum predictive power had been extracted from the 11 given features.

Metric	Optimized SVM (RBF)	Optimized KNN (Manhattan)
Accuracy	66.48%	66.40%
F1-Score	0.60	0.598
True Positives	312	309
True Negatives	509	511

- **Precision:** $\frac{TP}{TP+FP}$
- **Recall:** $\frac{TP}{TP+FN}$
- **F1-Score:** $2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$

The hard ceiling of ~66.5% aligns perfectly with established academic benchmarks for the COMPAS dataset, confirming that the remaining 33.5% error represents inherent human unpredictability uncaptured by the collected variables.

4.2 Error Disparity Analysis (The Fairness Audit)

While macro-level metrics indicated stability, partitioning the SVM's confusion matrix by race exposed severe algorithmic bias. The False Positive Rate (FPR) measures the percentage of innocent individuals incorrectly flagged as high-risk:

$$FPR = \frac{FP}{FP + TN}$$

- **African American FPR:** 32.89%

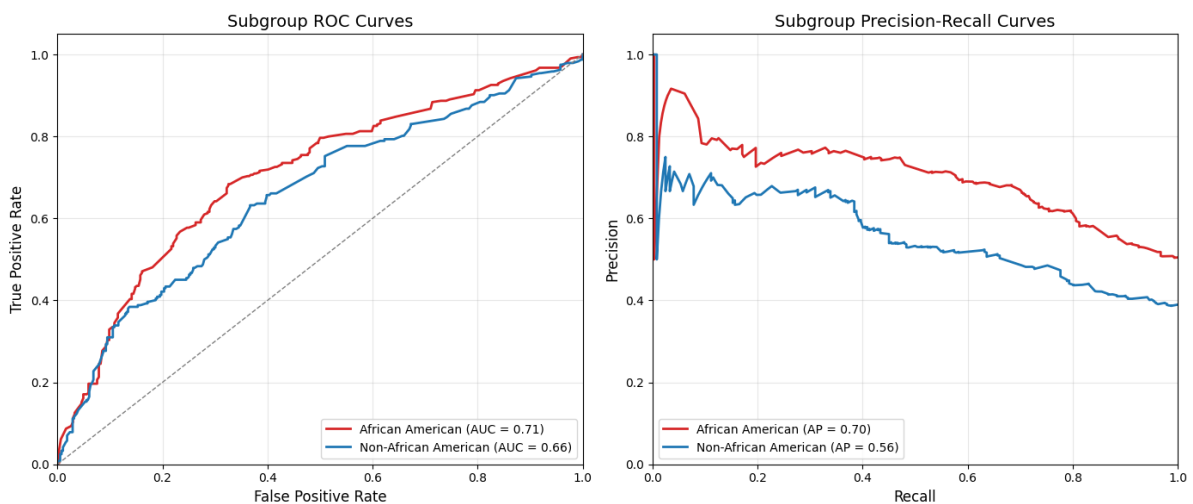
- **Non-African American FPR:** 19.53%
- **Subgroup ROC-AUC:** African American (0.71) vs. Non-African American (0.66).
- **Subgroup Precision-Recall (AP):** African American (0.70) vs. Non-African American (0.56).

The ROC and PR curves mathematically proved that the algorithm evaluated the two groups using fundamentally disparate risk landscapes across every possible probability threshold.

4.3 Bias Mitigation Strategies

Two distinct mitigation strategies were documented to address the FPR disparity:

1. **Fairness Through Unawareness (Feature Exclusion):** Dropping all explicit race features before training the SVM barely shifted the disparity (African American FPR remained at 31.91%, Non-African American FPR fell to 15.57%). This proved that explicit omission fails because surviving features act as proxies for race.
2. **Equalized Odds (Threshold Calibration):** By shifting the positive classification probability threshold independently for each group ($\tau_{AA} = 0.60$, $\tau_{Non-AA} = 0.42$), the False Positive Rates were mathematically forced into alignment (26.97% vs. 20.84%). This mathematically neutralized the predictive bias, reducing the disparity gap from 13.36% to just 6.13%.



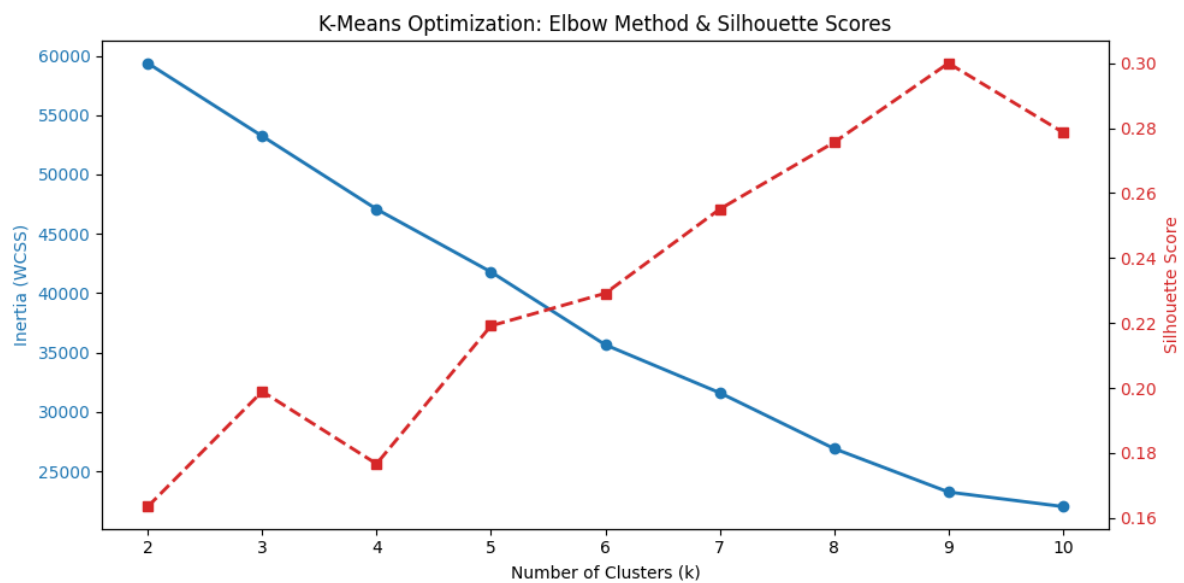
```
--- Optimized SVM (RBF) Results ---  
Best Parameters: {'C': 50, 'gamma': 'scale'}  
Overall Accuracy: 0.6648
```

Classification Report:

	precision	recall	f1-score	support
0	0.68	0.75	0.71	683
1	0.64	0.57	0.60	552
accuracy			0.66	1235
macro avg	0.66	0.66	0.66	1235
weighted avg	0.66	0.66	0.66	1235

Confusion Matrix:

```
[[509 174]  
 [240 312]]
```



```
--- Optimized KNN Results ---  
Best Parameters: {'metric': 'manhattan', 'n_neighbors': 25, 'weights': 'uniform'}  
Overall Accuracy: 0.6648
```

Classification Report:

	precision	recall	f1-score	support
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